Malkkeum Partner App — Product Spec

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Build the app that cleaning professionals (“pros”) use to take Malkkeum jobs, send quotes, run their schedule and team, complete jobs and get paid; it shares one backend with the customer app.

Overview

Malkkeum has three surfaces on one backend: the **customer app** (see the separate “Malkkeum App — Product Spec”), the **admin panel** for Malkkeum staff, and this **partner app** for pros. The UI language is **Korean**.

- **Who uses it:** independent cleaning professionals and small cleaning teams. A pro account is the team owner; owners can invite team members and assign jobs to them.
- **What pros do:** find paid jobs and accept them first-come, send quotes for move-out / move-in requests, see their schedule, contact customers through a masked number, check jobs complete with photos, and receive payouts.
- **What never happens here:** pros do not chat with customers. Customers talk only to the Malkkeum consultation team; pros talk only to Malkkeum operations. This keeps the customer app's single chat inbox unchanged.
- **All jobs are one-time.** There are no recurring jobs, subscriptions or memberships.

The prototype

The partner app prototype is the visual and behavioral reference; it opens already signed in as a sample pro with sample jobs, quotes and payouts.

- **Good test paths:** (1) Find jobs → a direct job → accept → it appears on Schedule with full address; (2) Find jobs → quote requests → send a quote → My quotes → open it → tap “[데모] 상담팀이 이 견적을 채택” → it moves to Schedule; (3) Schedule → today → job → “check job complete” → add photos and tick all items → a settlement appears in More → Settlements; (4) Chat tab → open the job thread with an unread badge, or Job detail → “말끔과 채팅하기”; (5) Schedule → the job marked “일정 변경 요청” → confirm; the cancelled job shows a banner and cannot be completed; completing the move-in job puts its payout in “잔금 대기”; (6) More → Services and areas → turn a service off and see its jobs disappear from Find jobs.
- **Reset:** More → 버전 정보 (version) restores the sample data.
- **Source:** one HTML file with plain JavaScript; ask for the file to read field names and rules.

What is fake in the prototype and must be built for real

Prototype behavior	Real requirement
Signed in as a sample pro, no onboarding	Sign-up, identity and bank verification, staff approval
Sample jobs stored in the browser	Jobs from the shared backend, real-time updates and push
Accept always succeeds	Atomic first-come accept; “already taken” error
Quote selected by a demo link	Staff select in the admin panel after the customer pays the deposit
“Call” shows a message	Masked 050 number connecting customer and pro
Photo tiles mark “added”	Camera / gallery upload, storage, shown to the customer
Settlement date computed in the browser	Settlement engine and real payouts
Operations chat replies are canned	Staff reply from the admin panel
Bank account and team are sample data	Verified account change, real invites by SMS

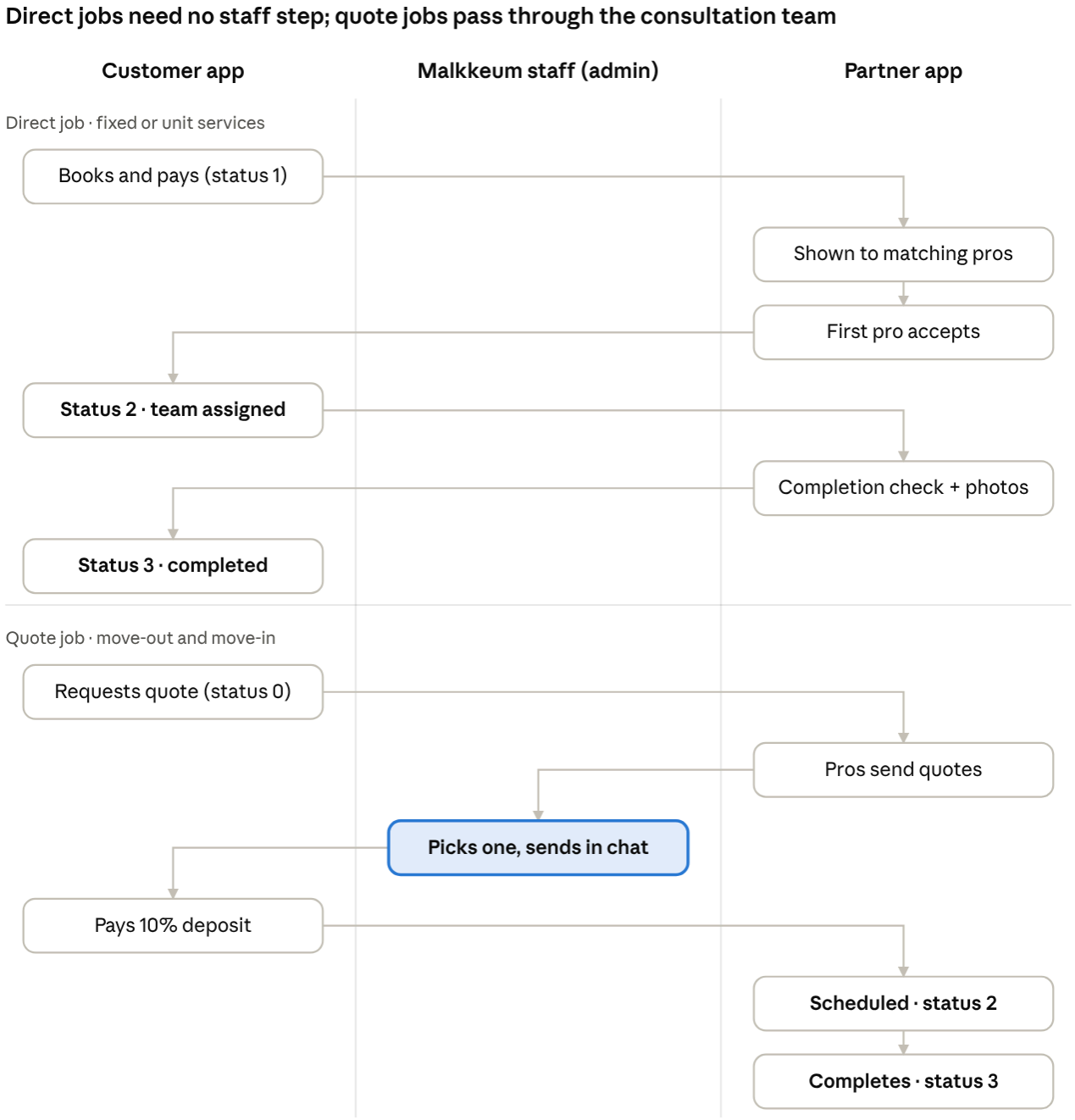
Scope

v1 covers the job marketplace, quoting, schedule, completion and payouts; anything not listed under “In” is out unless agreed in writing.

	Item
In	Pro sign-up with identity check, bank account and service areas (onboarding screens are not in the prototype)
In	5 tabs: Find jobs, My quotes, Schedule, Chat, More, and every screen in the Screens section
In	Accepting direct jobs, submitting / editing / withdrawing quotes
In	Team members and job assignment

	Item
In	Masked-number calling (customer ↔ pro)
In	Completion check with before / after photos and checklist
In	Settlement list and weekly payouts to the registered bank account
In	Chat tab collecting every conversation with Malkkeum operations (one thread per job plus general topics), push notifications
Out	Chat between pro and customer
Out	Recurring jobs, subscription plans, memberships
Out	Paid “cash” for bidding (pros pay a commission per completed job instead; see Decisions)
Out	Replying to reviews (button shown, feature later)

Job lifecycle



job lifecycle across the three surfaces

A direct job goes straight from the customer's payment to the pros. A quote job always passes through the consultation team, who choose one pro's quote and send it to the customer; the customer never talks to the pro.

Status mapping

One booking record drives both apps; the partner app shows its own job state, derived from the customer booking status plus the pro's actions.

Partner job state (Korean UI)	When	Customer booking status
Open — direct (바로 예약)	Customer paid a fixed / unit service; no pro yet	1 Confirmed
Open — quote request (견적 요청)	Customer submitted a move-out / move-in request	0 Consultation requested
Quote waiting (대기중)	Pro submitted a quote	0
Quote selected (채택)	Staff chose this quote and the customer paid the deposit	1 Confirmed, then 2 once the pro is attached
Quote not selected (미채택)	Staff chose another quote	unchanged
Quote closed (마감)	Request expired or was cancelled	cancelled or unchanged
Scheduled (on Schedule)	Pro accepted a direct job, or quote selected	2 Team assigned
Done	Pro submitted the completion check	3 Completed
Closed (마감)	Another pro accepted first, or the customer cancelled	—
Reschedule requested (일정 변경 요청)	Customer changed date/time after the pro was assigned; pro confirms or declines	2 with a pending reschedule request
Cancelled (취소됨)	Customer or staff cancelled an assigned job; stays on Schedule with a banner, cannot be completed	−1 Cancelled

Screens

The partner app has 5 bottom tabs and about 18 screens; use the prototype as the visual reference and this table as the checklist.

Area	Screen (prototype title)	Purpose	Key elements and actions
Tab 1 · Find jobs	업무찾기	Browse open jobs	Segment “direct” / “quote requests”, “show closed jobs” toggle, sort sheet (newest request, earliest visit, highest amount), job cards (service + type badge, NEW badge, date and slot, district, building, size, option or layout, payout or customer estimate), empty state
Find jobs	Job preview	Decide before taking a job	Hero with payout / estimate, schedule, space info, payout breakdown (customer price − fee); exact address and contact hidden; “accept job” (direct, with confirm sheet) or “send quote” (quote)
Find jobs	Send quote sheet	Bid on a quote request	Customer price input (pre-filled with the estimate), live payout preview, note to the consultation team, submit
Tab 2 · My quotes	내 견적 관리	Track submitted quotes	Status chips (all, waiting, selected, not selected, closed), “unseen results only” toggle, cards with my quote and customer estimate, NEW marker on unseen results, badge on the tab

Area	Screen (prototype title)	Purpose	Key elements and actions
My quotes	Quote detail	One quote	Status, my price, payout, estimate, submitted time, note, space info; “edit” and “withdraw” while waiting; a selected quote opens Job detail
Tab 3 · Schedule	스케줄 관리	Calendar of my team's jobs	Month view with prev / next, “N jobs” chip per day (dark = mine, light = a teammate's), “only mine” toggle, shortcuts to work-schedule settings and team
Schedule	Day sheet	Jobs on one day	Service badge with “cancelled” or “reschedule requested” badges, slot, masked customer name, assignee, address, “view”
Schedule	Job detail	Run an accepted job	Hero (service, payout, size, building, booking no.), schedule, customer (masked name, “call” via masked number, quiet hours note), assignee (change sheet), full address and size, service info, reschedule banner (old → new slot, “confirm” / “can't make it” which opens the job chat), cancelled banner, customer requests card (entry, parking, other), team-only memo, payout breakdown with “paid after the customer's balance” note for quote jobs, call / chat Malkkeum, “check job complete”

Area	Screen (prototype title)	Purpose	Key elements and actions
Schedule	Completion check	Close a job	Before photos (≥ 1), after photos (≥ 1), 4-item checklist, “complete” enabled only when all are done; success sheet with payout date
Tab 5 · More	더보기	Account hub	Profile (name, team), pending payout card, menus: settlements, reviews, services and areas, payout account, work schedule, team members, notifications, operations chat, notices, FAQ, version
More	Settlements	Payout history	Pending total, pending items with payout date, quote-job items marked “waiting for balance” (잔금 대기) until the customer pays, past payouts
More	Reviews	Customer ratings	Average stars on the menu, list of reviews, “reply” (later)
More	Services and areas	What jobs I see	Toggle per service (8), district chips for active areas
More	Payout account	Where money goes	Bank, masked account number, holder, “change” (needs re-verification)
More	Work schedule	When I work	Active weekdays, morning / afternoon toggles, days off
More	Team members	My team	Members list, invite sheet (name, phone), remove; owner badge

Area	Screen (prototype title)	Purpose	Key elements and actions
More	Notifications	Push preferences	New direct jobs, quote requests and results, operations chat, night-time quiet mode
Tab 4 · Chat	채팅 (chat list + chat room)	All conversations with Malkkeum operations in one place	List of threads sorted by latest message (job icon for job threads, last line, time, unread count), unread total as a tab badge, “new inquiry” sheet (pick a scheduled job or a topic: settlement, reschedule / cancel, app help, other); chat room with system lines, messages, typing indicator, “view job” link on job threads. “Chat with Malkkeum” on Job detail opens that job’s thread. FAQ and read-only profile live under More.

The prototype’s “[데모] 상담팀이 이 견적을 채택” (demo: staff select this quote) link and “version → reset demo data” are testing aids; in the real system staff select quotes in the admin panel.

Business rules

All rules are enforced on the server.

1. **Visibility:** a pro sees an open job only if its service is switched on in “Services and areas”, its district is in the pro’s active areas, and its date and slot fall inside the pro’s work schedule. The pro must be verified and active.
2. **Direct jobs are first-come:** the first pro to accept gets the job; accepting is atomic on the server, and other pros see the job as closed immediately. Accepting moves the customer booking to status 2.

3. **Quotes:** one quote per pro per request. A quote can be edited or withdrawn while waiting. Staff compare quotes in the admin panel, send the chosen price to the customer in chat, and when the customer pays the 10% deposit the quote becomes “selected” and the job appears on that pro′s schedule. Other quotes become “not selected”. A request with no selection by its visit date closes.
4. **Privacy before commitment:** before accepting or being selected, a pro sees only district, building type, size and space details. Full address, unit number and the customer′s masked name appear after. The customer′s real phone number is never shown; calls go through a masked (050) number valid until 3 days after the visit. Quiet hours: no calls after 20:00.
5. **Assignment:** the team owner can assign any job to a team member; the assignee gets a push. Only the owner sees payouts.
6. **Completion:** “complete” requires at least one before and one after photo and all checklist items. Photos are attached to the customer′s booking. Completion moves the booking to status 3 and notifies the customer.
7. **Fee and payout:** payout = customer price − platform fee (prototype: 20%). Payouts are batched weekly (prototype: the Wednesday after completion) to the verified bank account. Fee rate, schedule and tax handling are TBD.
8. **Cancellation by a pro:** allowed until 2 days before the visit via operations chat; later cancellations may carry a penalty (TBD). The job returns to the open list or to staff.
9. **Memo** on a job is visible to the pro′s team only, never to the customer.
10. **Chat threads:** one thread per job, created when staff or the pro first writes about it; staff use job threads for announcements (access codes, parking, schedule changes, quote selected). General threads have a topic. Customers never appear in partner chats.
11. **Customer requests:** entry method, parking and other notes the customer entered are shown on Job detail only after the pro accepted or was selected, and update live if the customer edits them.
12. **Reschedule requests:** when the customer changes the date of an assigned job, the job moves to the new slot with a “reschedule requested” banner. The pro confirms, or taps “can′t make it”, which posts in the job′s chat so staff can re-coordinate or reassign. Confirmation notifies the customer.
13. **Cancellations:** a cancelled job stays on Schedule with a cancelled banner and cannot be completed. Compensation for late cancellations (within 2 days) is TBD.
14. **Quote-job payouts:** a completed move-out / move-in job creates a settlement in “waiting for balance”; it becomes pending (and gets a payout date) only after the customer pays the balance in the customer app.

15. **Reviews:** written by customers in the customer app after completion; shown read-only in More → Reviews.

Data model additions

These entities extend the customer spec's data model; the prototype's source shows the same field names.

Entity	Key fields	Notes
Pro (team owner)	id, name, phone, verification status, active flag, team name, created_at	Login separate from customer accounts
TeamMember	id, pro_id, name, phone, role (owner / member), invite status	Members can be assignees, cannot see payouts
PartnerSettings	pro_id, services[] on/off, areas[] (district codes), weekdays[], slots[] (morning / afternoon), days_off[], notification prefs	Drives job visibility
Booking (extended)	+ pro_id, assignee_member_id, accepted_at, team_memo, completed_at; reads visit_notes, reschedule_request, cancelled_at, balance_paid from the customer side	Same record the customer app uses
Quote	id, booking_id, pro_id, amount, note, state (waiting / selected / rejected / closed), seen_by_pro, created_at, updated_at	Unique (booking_id, pro_id)
CompletionReport	booking_id, before_photos[], after_photos[], checklist[], submitted_by, submitted_at	Photos shown in the customer booking
Settlement	id, pro_id, booking_id, gross, fee, net, payout_date, state (waiting_balance / pending / paid / held)	Weekly payout batch
PayoutAccount	pro_id, bank, account number (encrypted), holder, verified_at	Changing it requires re-verification

Entity	Key fields	Notes
Review	booking_id, pro_id, rating 1–5, text, created_at	Written in the customer app (customer flow to be added)
OpsChatThread / Message	pro_id, booking_id (nullable), sender (pro / staff / system), body, created_at	Same chat service as customer consultation

Backend and admin panel additions

On top of the customer spec's backend and admin panel, the partner app needs the following.

- **Pro onboarding and verification:** sign-up, identity check, bank-account verification, optional training / approval step, activate / suspend a pro.
- **Job dispatch:** match open jobs to pros by service, area and schedule; push "new job" to matching pros; atomic accept; close a job for everyone else.
- **Quote desk (admin):** see all quotes per request side by side with the estimate, pick one, send it to the customer chat as a quote message, attach the pro when the deposit is paid; reject the rest.
- **Masked numbers:** a 050 virtual-number provider that pairs customer and pro per booking and expires after the job.
- **Photo storage:** upload and store completion photos, show them in the customer booking and the admin panel.
- **Settlement engine:** compute fee and net per completed job, weekly payout batch, payout file or bank-transfer API, hold / release, statements for tax.
- **Admin views:** pro list and profile, performance (acceptance, cancellations, ratings), reassign a job to another pro, penalties.
- **Operations chat:** a pro inbox next to the customer inbox in the admin panel.

Decisions needed

The client will settle these with you; please recommend an option and cost for each.

#	Decision	Options	Status
1	One codebase for both apps or two apps	Two store apps sharing code / one app with a role switch	TBD

#	Decision	Options	Status
2	Platform fee and payout schedule	Fee % per service; weekly / bi-weekly payout	TBD
3	Pro verification steps	ID check, business registration, training, interview	TBD
4	Masked-number provider	050 virtual-number vendors	TBD
5	Payout method	Bank-transfer API / manual payout file	TBD
6	Pro cancellation penalties	Rules and amounts	TBD
7	Who sees quote requests	All matching pros / a limited number per request	TBD
8	Tax handling for pros	Withholding for individuals / invoices for businesses	TBD
9	Customer review flow	Prototype: customer rates once after completion; moderation and hiding rules still open	TBD

Milestones and acceptance criteria

Three milestones, built alongside the customer app's milestones and accepted on real iOS and Android devices against staging.

1. **P1 — Onboarding and direct jobs** (with customer M1)
    - A pro signs up, is verified by staff, sets services, areas and schedule, and sees only matching jobs.
    - Two pros tap “accept” on the same job: exactly one gets it, the other sees it closed.
    - Accepting moves the customer's booking to “Team assigned”; the customer is notified.
2. **P2 — Quotes** (with customer M2)
    - Pros submit, edit and withdraw quotes; staff compare and select one in the admin panel.
    - After the customer pays the deposit, the selected pro sees the job on Schedule and the others see “not selected”.
3. **P3 — Run, complete and get paid** (with customer M3)

- Team owner invites a member and assigns a job; the member sees it with full address.
- Masked-number call connects customer and pro; real numbers never appear in either app.
- Completion with photos moves the booking to “Completed” and creates a settlement; the weekly batch pays the verified account and the item shows as paid.

Glossary

Customer-side terms (평, 원룸, 이사 청소, 입주 청소, 오피스텔, 빌라) are in the customer spec's glossary.

Korean	Meaning
프로 (pro)	A cleaning professional or team owner using the partner app
바로 예약	Direct job: customer already paid a fixed / unit service; first pro to accept gets it
견적 요청	Quote request: move-out / move-in job that needs a price
채택 / 미채택	Quote selected / not selected
마감	Closed (taken by someone else, expired or cancelled)
담당자	Assignee: the team member doing the job
정산	Settlement / payout to the pro
수수료	Platform fee
안심번호 (050)	Masked phone number that hides real numbers
전용면적	Exclusive floor area, the size used for jobs